

imaging techniques before VCE in suspected Crohn's disease^[22]. However, in 2015 based on broader evidence, the European Society for Gastrointestinal Endoscopy (ESGE) recommended not using cross sectional imaging or patency capsule before VCE in patients with suspected Crohn's disease in the absence of obstructive symptoms. In contrast, in established Crohn's disease, imaging techniques and patency capsule are recommended to precede VCE^[23].

Patients with a small bowel (SB) tumor seem to have a slightly higher risk of retention. A suspected tumor as an indication for VCE was associated with an odds ratio of 3.9 (95%CI: 1.2-12.8, $P = 0.026$)^[17]. However, clinical symptoms of such tumors are typically bleeding or iron deficiency anemia. As tumors only present in a small subgroup of patients presenting with bleeding/anemia, retention even in this subgroup is rare, mostly asymptomatic, and diagnostic rather than a complication, ESGE recommends against routine precautions tests before VCE in bleeding patients. However, if a tumor is suspected by imaging techniques, device assisted enteroscopy with the option of obtaining histology is preferred over VCE^[23].

In sum, suspected or known GI stenosis is a contraindication unless intestinal patency is proven, best by the passage of an intact patency capsule. The risk for capsule retention should be assumed in patients with known Crohn's disease, clinical or radiologic signs of obstruction, a history of abdomino-pelvic radiation, and after small bowel resection. Patients undergoing VCE for mid-GI bleeding without the above risks do not require preceding radiology or a patency capsule.

VCE IN PATIENTS WITH MOTILITY DISORDERS

VCE is not indicated for the diagnosis of GI motility disorders. For this purpose, a specifically designed, non-imaging wireless motility capsule (SmartPill, Medtronic plc, Dublin, Ireland) has been developed. Data from sensors measuring pH, pressure, and temperature are transmitted wirelessly for up to 5 d allowing diagnosis of gastroparesis, and prolonged transit times in the small bowel, colon or combined disorders^[24-27].

Nevertheless, standard video capsule was applied in 18 patients with chronic intestinal dysmotility in the search for associated mucosal lesions. Three capsules were retained in the stomach for > 2 h, one of them during the entire recording time. However, no permanent retention, symptoms, or need for interventional treatment occurred^[28]. Another study included 36 patients with severe symptomatic intestinal motor disorders for analysis of VCE image patterns compared with controls. No adverse events were mentioned in this report^[29].

Although indication of VCE for diagnosis of GI motility disorders has yet to be considered as

experimental, known or yet undiagnosed motility disorders may jeopardize routine VCE performed for other indications. Prolonged esophageal or gastric passage may lead to incomplete visualization of the small bowel, *i.e.*, the cecum is not reached during working capacity of the batteries. Moderate prolongation seems to be compensated by longer battery life span in newer capsule generation^[30].

VCE systems using an external recorder have the ability to display transmitted images in real-time during the procedure^[31-33]. Significantly prolonged gastric transit time can be identified by this real-time viewer and a prokinetic agent can be administered^[34]. A single center study reported a higher completion rate and diagnostic yield when a real time viewer was used and the capsule was placed endoscopically into the duodenum in the case of prolonged gastric transit time (> 60 min)^[35]. The unselected primary endoscopic placement of the capsule into the duodenum to circumvent possible gastroparesis had no effect on complete small bowel visualization in a single center analysis of 687 hospitalized or out-patients compared to swallowing the capsule^[36]. In a prospective single-center study including 100 VCE studies, a pathologic Gastroparesis Cardinal Symptoms Index questionnaire could not predict a prolonged gastric transit time nor did a delayed gastric passage have any clinical significance^[37].

GI motility disorders are no contraindication for VCE. The routine use of a real time viewer directly after swallowing the capsule and after an hour enables detection of aspiration (see below) and esophageal or gastric retention and consecutive intervention.

VCE IN PATIENTS WITH IMPLANTABLE CARDIAC DEVICES

The radio transmitters of the first capsule endoscopes work with a carrier frequency of 434.1 MHz in PillCam and 433.8 MHz in EndoCapsule, similar to the C-Net mobile cellular system (450 MHz). The frequency in the newly available OMOM Capsule is 2.4 GHz. Two studies revealed electromagnetic interference (EMI) between cardiac pacemakers (PM) and the C-Net mobile cellular system in 22.4%-30.7% of the tested pacemakers^[38,39]. However, the radiated power of C-Net mobile phones with 2 W is several factors higher than that of VCE with max. 100 nW. EMI with implantable cardiac devices at 2.4 GHz was also investigated in two studies^[40,41] showing no risk of interference. Nevertheless, users of VCE estimated EMI between capsules and cardiac devices possibly being life-threatening for patients. Since the introduction of VCE, several *in vitro* and *in vivo* studies analyzed EMI between VCE (PillCam and EndoCapsule) and PMs (*in vitro*:^[42-44], *in vivo*:^[44-54]), implantable cardioverter defibrillators (ICD) (*in vitro*:^[55,56], *in vivo*:^[45-48,54,56-59]) and left ventricular assist devices (LVAD) (*in vitro*: